

On Cofactuals

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Abstract

The distribution of policy gains depends on how outcomes are paired across regimes. I study a shared-component model in which policy labels fix the loadings of independent interaction terms. Under local exponential integrability, the marginal laws admit this model if and only if their Möbius-transformed cumulant generating functions are admissible. The same transforms then recover the component laws and the joint law within the class, including every policy-gain distribution. A modular Bellman model gives an economic foundation, with reoptimization illustrating its scope. Moment versions give marginal variance inequalities and paired covariance equalities. The observation scheme determines which restrictions can be assessed: the sampling results cover selected paired and state-adjusted covariance moments. A component-covariance bound and analytical examples connect departures from orthogonality to gain dispersion. Simulations examine the calibration of the paired tests.

JEL Codes: C10, C18, C61.

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1. Introduction

How dispersed are the gains from a policy? To study this question, consider outcomes for the same population under a baseline and an alternative. Their pairing matters for the dispersion of individual gains. For baseline outcome $Y_\emptyset$ and policy outcome Y_1 ,

$$(1) \quad \text{Var}(Y_1 - Y_\emptyset) = \text{Var}(Y_1) + \text{Var}(Y_\emptyset) - 2 \text{Cov}(Y_1, Y_\emptyset).$$

The two variances can be learned from marginal information. The covariance depends on which outcomes belong to the same unit or structural draw. Thus the same marginal laws can support different accounts of gain heterogeneity (Firpo and Ridder 2019; Gao et al. 2025).

I study a shared-component restriction on this pairing. The two-regime case is a useful place to begin. Write $Y_\emptyset = C_\emptyset$ and $Y_1 = C_\emptyset + C_1$, with the baseline component $C_\emptyset$ independent of the increment C_1 . If their moment generating functions are finite near zero, the marginal cumulant generating functions satisfy

$$(2) \quad K_{C_1}(t) = K_1(t) - K_\emptyset(t).$$

The condition has a simple interpretation: the difference on the right must be the cumulant generating function of an increment law. When it is, that law gives the distribution of the gain under the maintained independence restriction. For Gaussian marginals with means $m_\emptyset, m_1$ and variances $v_\emptyset, v_1$, compatibility reduces to $v_1 \geq v_\emptyset$. The gain law is then $N(m_1 - m_\emptyset, v_1 - v_\emptyset)$, including a degenerate increment at equality. The chosen baseline fixes the direction of this restriction.¹

With several policy switches, each subset labels a regime. I call the resulting baseline-anchored interaction terms *cofactual components*.² For two switches, $C_i = Y_i - Y_\emptyset$ and $C_{12} = Y_{12} - Y_1 - Y_2 + Y_\emptyset$. Möbius inversion writes each outcome as the sum of the components whose labels it contains. The policy labels fix the loadings across regimes. Independence then supplies the restriction on cross-world dependence.

Main result. Theorem 1 gives a marginal-to-joint characterization. Under local exponential integrability, the marginal laws admit an independent-component coupling if and only if every Möbius-transformed CGF is the CGF of a probability law. When the condition holds, those transforms construct the unique component laws and the unique joint law within the factorized class. Corollary 1 then gives every policy-gain distribution. The

¹Reversing the baseline applies the same construction with the regime roles exchanged and can give a different compatibility condition.

²The label refers to baseline-anchored Möbius interaction terms. Whether a regime is factual or counterfactual depends on the observation design and the reference case.

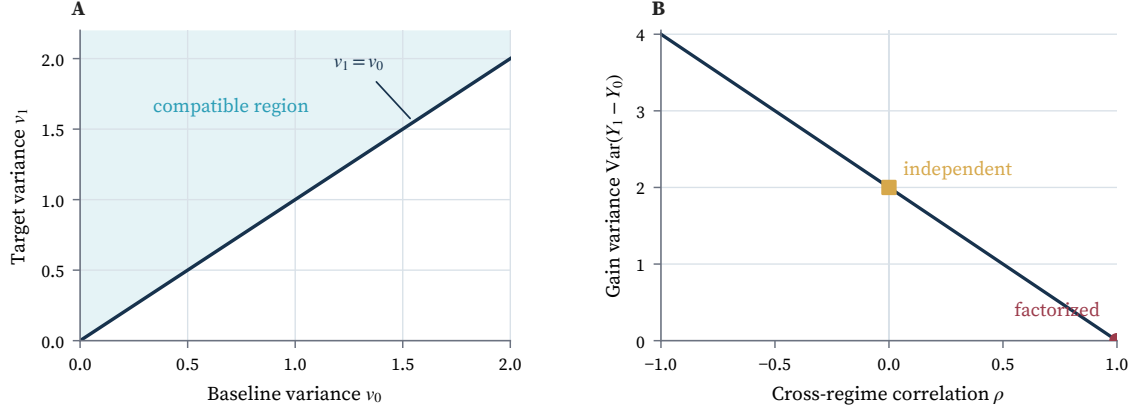


FIGURE 1. Marginal compatibility and coupling-sensitive gain dispersion

Panel A gives the factorized Gaussian compatibility region for baseline variance v_0 and target variance v_1 ; the diagonal gives a degenerate increment. Panel B holds both marginal variances at one and plots $\text{Var}(Y_1 - Y_0) = 2(1 - \rho)$ against the cross-regime correlation. The factorized equal-variance model corresponds to $\rho = 1$, while an independent pairing corresponds to $\rho = 0$. The comparison is algebraic.

theorem therefore links an observable collection of marginal laws, a latent component representation, and a constructive joint-law formula. For example, when both regime marginals are standard normal, the factorized model makes the gain zero almost surely. An independent pairing of those same marginals gives gain variance two. The marginal evidence is the same; the maintained coupling determines the gain distribution.

It is useful to ask which economic primitives support the restriction. Under a common policy-evaluation operator, independent reward modules remain independent after propagation through a fixed Bellman resolvent. Reoptimization can instead make the baseline and increment dependent. A common pre-intervention state can also account for shared observable heterogeneity, with the restriction applied within each state.³

The distributional result also has lower-order implications. At second order, orthogonality of components is equivalent to

$$(3) \quad \text{Cov}(Y_S, Y_T) = \text{Var}(Y_{S \cap T}).$$

The intersection records the components shared by the two regimes. Under orthogonality, the variance surface has nonnegative Möbius coefficients; higher cumulants satisfy corresponding intersection identities. For a general coupling, a union identity shows which component covariances enter a marginal variance contrast. The variance-error bound and reward-credit calculation trace their effect on the policy target.

³Conditional independence within the recorded state permits unconditional dependence generated by variation in that state.

Observation and assessment. The marginal reconstruction is a population result. Separate regime samples can supply its inputs under an appropriate assignment and support design. Paired rows also record how outcomes are joined and permit assessment of selected restrictions on that joint law. The sampling results below concern covariance equalities and state-adjusted residual moments for a fixed intervention family. Their calibration uses the recorded pairing.⁴ Estimation of the reconstructed gain distribution and boundary-calibrated tests of marginal compatibility are further statistical questions. The present tests address the paired moment restrictions.

Related literature. Chernozhukov et al. (2013) develop inference for marginal counterfactual distributions. Here those distributions are the population inputs, and the object to recover is their cross-regime joint law. Gao et al. (2025) optimize counterfactual functionals over marginal-compatible couplings, while Wu and Mao (2026) use variation across experiments with transportability and rank conditions to identify joint potential-outcome distributions. The present paper studies a specified alternative source of identifying information: independence of interaction components whose loadings are fixed by policy labels. Theorem 1 characterizes the marginal inputs that this restriction admits and the joint law it selects.

Compatibility of marginal objects also arises outside counterfactual analysis. Wang (2004) use Lancaster interactions to characterize completion of overlapping marginal densities by an all-positive joint density. Azrieli and Rehbeck (2026) characterize consistency of marginal menu and choice frequencies through the core of a totally monotone game and show that random utility adds no restriction on their unrestricted marginal dataset. These results concern feasibility of their respective marginal systems. The factorized class here fixes Boolean component loadings and turns admissible cumulant transforms into a unique joint law within that class.

McKay and Wolf (2023) ask which causal responses suffice for policy-rule counterfactuals in linearized structural models. Their policy-rule problem and the fixed-regime pairing problem here use different observed objects and maintained restrictions. Christensen and Connault (2023) bound counterfactuals over neighborhoods of latent distributions subject to model constraints. The variance comparison here fixes the actual component variances and bounds the contribution of component covariance. Its reference and information requirements are stated with the bound.

The coordinates are classical Möbius inversion, related to Harsanyi dividends, factorial contrasts, and causal or attribution decompositions (Rota 1964; Harsanyi 1963; Dasgupta et al. 2015; Jansma 2025; Forré and Jansma 2025; Gao and Zhao 2025). Shared-source and convolution constructions provide related probabilistic representations (Schulz et al. 2021;

⁴Examples are a simulator’s common draw and an empirical pairing whose unit-level interpretation is supported by the observation design.

Balan 2002); the covariance here is a Boolean meet matrix (Mattila and Haukkanen 2014). Mesters and Zwiernik (2024) study identification of a mixing matrix using restrictions on component moments and cumulants.⁵ Möbius inversion and product construction supply the algebra; the analysis uses them to connect marginal admissibility, policy-gain laws, and the restrictions available under different observation schemes.

For assessment, I use covariance-testing and semiparametric methods (Shah and Peters 2020; Scheidegger et al. 2022; Chernozhukov et al. 2018, 2022) for the moments implied by the coupling restriction. The online appendix also studies precision when paired evaluations are costly, taking the diagnostic and required world bundle as given.

The paper moves from cofactual coordinates to marginal reconstruction, a dynamic foundation, moment restrictions, and observation design. A reward-credit example carries the same objects through policy-gain dispersion. Supporting proofs, inference details, and numerical exhibits appear in the online appendix.

2. Cofactual coordinates

Fix a baseline and $p \geq 1$ labeled interventions. Let $\mathcal{W} = 2^{[p]}$, ordered by inclusion, and write $n = 2^p$ for the number of regimes. Each set denotes a joint regime with a fixed meaning throughout the family. We use Y_i for $Y_{\{i\}}$ and Y_{12} for $Y_{\{1,2\}}$; two-regime examples also write $Y_0 = Y_\emptyset$. History-dependent interventions require a richer index.

DEFINITION 1 (Counterfactual system). *On a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a finite counterfactual system is a family*

$$Y = \{Y_S : S \in \mathcal{W}\},$$

where each $Y_S : \Omega \rightarrow \mathbb{R}^d$ is measurable. The coordinate $Y_\emptyset$ is the reference target. When the actual regime is the baseline, it is the factual outcome; the remaining coordinates are counterfactual outcomes relative to that regime.

The target may be a potential outcome, policy value, equilibrium moment, or welfare vector. The common ω is the unit, type, or structural draw held fixed across regimes. It specifies what it means to compare outcomes for the same underlying case. A simulation implements the resulting coupling; an empirical application gives it content through the model and observation design.

⁵Their loading matrix is an unknown object. Here the intervention labels fix the loading matrix before the component restrictions are imposed.

DEFINITION 2 (Cofactual component). For $A \subseteq [p]$, define

$$(4) \quad C_A = \sum_{B \subseteq A} (-1)^{|A|-|B|} Y_B.$$

For $A \neq \emptyset$, C_A is the cofactual component associated with A . The coordinate $C_\emptyset = Y_\emptyset$ is the reference component.

PROPOSITION 1 (Cofactual representation). For every $S \subseteq [p]$,

$$(5) \quad Y_S = \sum_{A \subseteq S} C_A.$$

The collection $\{C_A\}_{A \subseteq [p]}$ is the unique collection with this property.

PROOF. Substitution and a change in the order of summation give

$$\sum_{A \subseteq S} C_A = \sum_{B \subseteq S} Y_B \sum_{A: B \subseteq A \subseteq S} (-1)^{|A|-|B|}.$$

Writing $A = B \cup D$, the inner sum is $(1 - 1)^{|S \setminus B|}$. Only $B = S$ remains. Uniqueness follows by applying the same inversion formula to any representation of the form (5). $\square$

Order the worlds consistently with inclusion and stack Y_S and C_A . Let

$$M_{A,B} = (-1)^{|A|-|B|} \mathbf{1}\{B \subseteq A\}, \quad Z_{S,A} = \mathbf{1}\{A \subseteq S\}.$$

Then

$$(6) \quad C = MY, \quad Y = ZC, \quad MZ = ZM = I.$$

For vector targets, the stacked transform is $M \otimes I_d$, with inverse $Z \otimes I_d$. I suppress this identity factor in the scalar notation below. The original outcomes can be recovered from the components, so the change of coordinates retains the full system.

The coordinates also define a multilinear polynomial, $\mathcal{Y}_\omega(z) = \sum_A C_A(\omega) z^A$. Its values at Boolean vertices recover the regimes, and its mixed derivatives at zero recover the components. Interior values interpolate algebraically between regimes; interpreting them as randomized policies requires a randomization design. Under a degree restriction, the same representation yields compression formulas indexed by the number of policy switches in an interaction. The cumulant order describes a feature of its probability law.

3. Marginal compatibility and policy-gain distributions

The coordinate representation holds for every joint law. The question now is which marginal distributions admit a joint law with independent components. Throughout this distributional analysis, the target is scalar.

DEFINITION 3 (Cofactual factorization). *A scalar counterfactual system is cofactually factorized when the family*

$$\{C_A : A \subseteq [p]\}$$

is mutually independent.

Under factorization, each C_A is a latent source shared by the worlds containing A . The root component enters every world. Independence is the substantive restriction on these sources and can be imposed conditional on a common state.

Factorization implies a marginal convolution identity with component laws $\mu_A = \mathcal{L}(C_A)$ such that

$$(7) \quad \mathcal{L}(Y_S) = \bigstar_{A \subseteq S} \mu_A.$$

This is a Boolean convolution system. The representation is close in spirit to set-indexed processes with independent increments (Balan 2002). Here the set index is a policy world, and the component laws will be used to recover cross-world dependence from separate world distributions.

REMARK 1 (Reference regime). *Factorization is indexed by the chosen factual root. With one intervention and square-integrable independent $C_\emptyset, C_1$, reversing the reference regime gives root $Y_1 = C_\emptyset + C_1$ and increment $-C_1$. Their covariance is $-\text{Var}(C_1)$. Thus the direction of the intervention comparison belongs to the model specification. Relabeling interventions while retaining the root preserves the restriction.*

3.1. Marginal-to-joint characterization

ASSUMPTION 1 (Local exponential integrability). *The world marginal MGFs are finite on a common open interval containing zero.*

DEFINITION 4 (Admissible CGF). *A real-valued function on a neighborhood of zero is an admissible CGF when it is the cumulant generating function of a probability law with a finite MGF on that neighborhood.*

THEOREM 1 (Marginal-to-joint characterization). *Let $P = (P_S)_{S \in \mathcal{W}}$ be a family of scalar marginal laws satisfying Assumption 1, with CGFs K_S . Define*

$$(8) \quad H_A(t) = \sum_{B \subseteq A} (-1)^{|A|-|B|} K_B(t).$$

A factorized coupling of P exists if and only if every H_A is admissible on a common neighborhood of zero. Within this class, the component laws and joint law are unique. On their respective neighborhoods,

$$(9) \quad K_S(t) = \sum_{A \subseteq S} H_A(t),$$

$$(10) \quad K_Y^{\text{joint}}(t) = \sum_{A \subseteq [p]} H_A \left(\sum_{S \supseteq A} t_S \right).$$

Local transform domains. If the n marginal MGFs are finite for $|t| < \delta$, then for any compatible coupling and $\max_S |t_S| < \delta/n$, Hölder's inequality gives

$$\mathbb{E} \exp \left(\sum_S t_S Y_S \right) \leq \prod_S \{ \mathbb{E} \exp(n t_S Y_S) \}^{1/n} < \infty.$$

Thus every compatible joint law has a local joint MGF. A finite linear map, including M , retains this property on a smaller neighborhood.

PROOF. Suppose a factorized coupling exists. Since $Y_S = \sum_{A \subseteq S} C_A$, independence gives

$$\mathbb{E}[e^{t Y_S}] = \prod_{A \subseteq S} \mathbb{E}[e^{t C_A}].$$

Taking logs gives (9); Boolean inversion gives (8). The local-domain argument above makes each component transform finite where used. Each H_A is therefore admissible.

Conversely, choose the probability law μ_A with local CGF H_A and construct the independent family $C_A \sim \mu_A$ on the finite product space. Set $Y_S = \sum_{A \subseteq S} C_A$. Its CGF is the sum in (9), which equals K_S by inversion. Local MGF uniqueness gives the marginal law P_S . The same uniqueness identifies every μ_A . Their product law is unique, and $Y = ZC$ determines its image. Finally,

$$\sum_S t_S Y_S = \sum_A C_A \sum_{S \supseteq A} t_S$$

proves (10) by independence. All identities are on a common neighborhood small enough for their arguments. This proves existence, reconstruction, and uniqueness within the stated class. $\square$

Möbius inversion gives a candidate H_A for any marginal CGFs.⁶ Admissibility asks whether that candidate is the CGF of a probability law. Normalization at zero and nonnegative second derivatives provide necessary checks. In general, the answer depends on the full local function; for Gaussian marginals it reduces to parameter restrictions.

What the conditions provide. The fixed baseline and complete intervention family determine the invertible incidence matrix. Admissibility provides a probability law for each transformed CGF, and local exponential integrability makes these transforms finite and law-determining near zero. Independence joins the recovered component laws by their product. Degenerate components are included. Each condition thus has a separate role in the construction.

These transform conditions belong to the full-law result. The variance characterization uses finite second moments and orthogonality; the paired Wald result uses fourth moments and a nonsingular influence covariance. The latter condition concerns sampling calibration. A factorized law can be well defined even when that covariance is singular.

3.2. Joint-law identification and policy gains

COROLLARY 1 (Marginal-to-joint identification). *Suppose Assumption 1 and the admissibility condition in Theorem 1 hold. Within the factorized class, the marginal laws identify all component laws, all cross-world joint cumulants, and the full joint law in (10). For regimes S, T , define*

$$(11) \quad d_A(S, T) = \mathbf{1}\{A \subseteq S\} - \mathbf{1}\{A \subseteq T\}.$$

The policy gain $G_{S,T} = Y_S - Y_T$ has local CGF

$$(12) \quad K_{G_{S,T}}(t) = \sum_{A \subseteq [p]} H_A(d_A(S, T)t),$$

which uniquely determines its probability law.

PROOF. Theorem 1 identifies the component and joint laws. Equation (5) expresses the gain as $\sum_A d_A(S, T)C_A$. Independence gives the sum of CGFs in (12). Local exponential integrability and MGF uniqueness determine the gain law. Components entering both regimes have zero coefficient and cancel. $\square$

⁶For Möbius inversion on finite partially ordered sets, see Rota (1964); Chateauneuf and Jaffray (1989) develops its use for capacities and lower probabilities. Here inversion is applied to cumulant-generating functions, and admissibility supplies the probabilistic restriction.

Equation (12) gives the economic content of the reconstruction. Within the maintained class, it determines gain probabilities, quantiles, and dispersion. For a baseline comparison $T = \emptyset$, the gain consists of the components indexed by nonempty subsets of S . Between two arbitrary regimes, components specific to S enter positively and those specific to T enter negatively. Shared components cancel.

4. Economic foundations and their scope

Independent reward modules provide one economic setting for factorization. A common decision rule propagates the modules separately; reoptimization can link the resulting components. The two analytical examples below follow this change while keeping track of the latent draw and transition rule.

4.1. Modular Bellman propagation

Let the state space be finite and let σ be a stationary policy. Write P_σ for its transition matrix and assume $\beta \in (0, 1)$. For each $A \subseteq [p]$, let u_A be a random reward vector, with the collection $\{u_A\}$ mutually independent. Randomness here indexes latent types or reward coefficients across evaluations; each Bellman equation evaluates the realized reward vector. The rule σ and matrix P_σ are fixed across these draws. Suppose the reward vector under world S is

$$(13) \quad r_S^\sigma = \sum_{A \subseteq S} u_A.$$

The policy value solves

$$(14) \quad V_S^\sigma = r_S^\sigma + \beta P_\sigma V_S^\sigma.$$

PROPOSITION 2 (Modular Bellman propagation). *Under (13)–(14), the value cofactual associated with A is*

$$(15) \quad C_A^V = (I - \beta P_\sigma)^{-1} u_A.$$

Hence the random vectors $\{C_A^V\}$ are mutually independent. For any fixed linear welfare functional $q^\top V_S^\sigma$, the resulting scalar counterfactual system is cofactually factorized. If this fixed σ is optimal in every world almost surely, the same statements hold for the optimal value family.

PROOF. The Bellman resolvent gives

$$V_S^\sigma = (I - \beta P_\sigma)^{-1} \sum_{A \subseteq S} u_A = \sum_{A \subseteq S} (I - \beta P_\sigma)^{-1} u_A.$$

Uniqueness of the Möbius representation identifies the A th value cofact with (15). Separate measurable linear images of mutually independent random vectors remain mutually independent. A fixed linear functional preserves independence across the indexed components. $\square$

The same argument applies conditional on X when $\sigma(X)$ and $P_\sigma(X)$ depend on the common state alone and the u_A are independent given X . The value components are then conditionally independent. Choosing the rule using the realized reward modules introduces an additional dependence channel that requires its own analysis.

The common resolvent is doing the work: each value component depends on its own reward module. A change in the optimal rule, the transition matrix, or dependence among the primitive modules can change the component covariance. Several such effects may offset one another in a particular moment. To interpret a policy change that affects behavior, the analyst therefore needs the decision problem under each regime (Lucas 1976; Sargent and Stachurski 2026).

4.2. Reoptimization and stochastic dependence

A one-state example shows a boundary of the common-policy condition. Let U take -1 and 1 with equal probability, and let the agent choose $a \in \{0, 1\}$. Baseline flow payoff is aU ; after an intervention it is $a(U - 2)$. Holding U fixed across regimes, reoptimization gives

$$Y_0 = \frac{\max\{0, U\}}{1 - \beta}, \quad Y_1 = 0, \quad C_1 = -C_\emptyset.$$

It follows that

$$(16) \quad \text{Cov}(C_\emptyset, C_1) = \text{Var}(Y_1) - \text{Var}(Y_0) = -\frac{1}{4(1 - \beta)^2}.$$

Here the negative covariance violates the paired orthogonality equality, and the fall in marginal variance violates the compatibility inequality. The change comes from reoptimization with the same latent draw and transition rule.

5. A policy-gain illustration

Consider a one-state discounted policy-evaluation problem with $\beta = 0.95$. A persistent pre-intervention group $X \in \{-1, 1\}$ has equal probabilities. Let U, V be independent standard normals, also independent of X . A reward credit changes the persistent flow payoff from

r_0 to r_1 :

$$(17) \quad \begin{aligned} r_0 &= 2 + X + U, \\ r_1 - r_0 &= a(X) + \sigma(X)\{\rho U + \sqrt{1 - \rho^2}V\}, \end{aligned}$$

where $a(X) = 0.5 + 0.25X$, $\sigma(-1) = 1$, $\sigma(1) = 3$, and $|\rho| < 1$. Each draw specifies persistent reward heterogeneity for the evaluation horizon. Write $Y_s = (1 - \beta)V_s = r_s$ for the annuity value. The declared coupling holds (X, U, V) fixed across regimes. The credit increment is $C_1 = Y_1 - Y_0$ and the root is $C_\emptyset = Y_0$.

Policy question. The target is the dispersion of gains, $\tau = \text{Var}(Y_1 - Y_0)$. Direct calculation gives

$$(18) \quad \tau = \mathbb{E}[\sigma(X)^2] + \text{Var}\{a(X)\} = 5 + 1/16 = 5.0625.$$

Under $\rho = 0$, the components are independent conditional on X . Their unconditional covariance is $\text{Cov}(2 + X, 0.5 + 0.25X) = 0.25$. Consequently the unadjusted marginal variance difference is 5.5625, while the state-adjusted identity gives

$$(19) \quad \mathbb{E}[\text{Var}(Y_1 | X) - \text{Var}(Y_0 | X)] + \text{Var}\{\mathbb{E}[C_1 | X]\} = 5.0625.$$

The 0.5 gap in the unadjusted calculation comes from common-state mean variation.

Sensitivity to the coupling. For general ρ , the residual component covariance is $\gamma(X) = \rho\sigma(X)$. The same proxy on the left of (19) becomes $5.0625 + 4\rho$, while the gain variance in (18) stays fixed. At $\rho = 0.2$, the proxy is 5.8625, exceeding the target by 0.8. The union identity explains the difference: it is $2\mathbb{E}[\gamma(X)]$. Thus $\theta = \mathbb{E}[R_\emptyset R_1] = 2\rho$ assesses the covariance restriction used by the proxy. Along this sensitivity path, the marginal law of Y_1 also changes with ρ .

Assessing the variance formula. A simulator records X, Y_0, Y_1 for each common latent draw. Subtraction recovers C_1 , and a cross-fitted residual product estimates θ . At $\rho = 0$, the signal has conditional variances 1 and 9 in the two groups. It diagnoses the covariance correction; the economic target remains $\tau = \text{Var}(Y_1 - Y_0)$.

The acquisition exercise fixes a prescribed bundle and expected budget, and the feasible-estimator experiment evaluates the resulting diagnostic.⁷ Behavioral or equilibrium responses belong in the regime-specific outcome maps.

⁷An acquisition rule optimized for τ would use the influence function of that policy target.

6. Moment restrictions, common states, and variance error

The full-law result can also be read through moments. Marginal variances and cross-regime cumulants give lower-order restrictions, and common-state adjustment helps locate the covariance terms relevant to a policy-contrast variance.

6.1. Intersection restrictions and policy-gain variance

Full factorization implies a sequence of finite-order restrictions.

DEFINITION 5 (Finite-order cofactual orthogonality). *Fix an integer $q \geq 2$. A counterfactual system is q -cofactually orthogonal when, for every $2 \leq r \leq q$,*

$$(20) \quad \text{Cum}(C_{A_1}, \dots, C_{A_r}) = 0$$

whenever at least two of the indices $A_1, \dots, A_r$ differ. The case $q = 2$ is called cofactual orthogonality.

THEOREM 2 (Intersection-cumulant characterization). *Suppose all joint moments through order q exist. The system is q -cofactually orthogonal if and only if, for every $2 \leq r \leq q$ and every $S_1, \dots, S_r \in \mathcal{W}$,*

$$(21) \quad \text{Cum}(Y_{S_1}, \dots, Y_{S_r}) = \kappa_r(Y_{S_1 \cap \dots \cap S_r}).$$

If the joint MGF of Y exists in a neighborhood of the origin, then the system is cofactually factorized if and only if (21) holds for every $r \geq 2$.

Proof. See Appendix C.1.

Shared components generate the intersection in the forward implication. The reverse implication applies Möbius inversion to each cumulant index. Thus the same restriction has a component form and a world form.

COROLLARY 2 (Meet covariance and cofactual orthogonality). *For a square-integrable scalar system, the following are equivalent:*

1. *the cofactual components are pairwise uncorrelated;*
2. *for all $S, T \in \mathcal{W}$,*

$$(22) \quad \text{Cov}(Y_S, Y_T) = \text{Var}(Y_{S \cap T}).$$

For the policy gain in Corollary 1, the second-order implication is

$$(23) \quad \text{Var}(G_{S,T}) = v(S) + v(T) - 2v(S \cap T), \quad v(S) = \text{Var}(Y_S).$$

TABLE 1. A hierarchy of cofactual restrictions

Layer	Restriction in cofactual coordinates	Implication in world coordinates
Unrestricted	none	arbitrary cross-world coupling
Second order	$\text{Cov}(C_A, C_B) = 0$ for $A \neq B$	meet covariance
Order q	mixed cumulants vanish through q	intersection cumulants through q
Full factorization	$\{C_A\}$ mutually independent	CGF inversion and joint-law recovery

The rows are nested when the required moments exist. Under a local joint MGF, validity of the finite-order restrictions at every order reaches the full factorization row.

Pairwise orthogonality and finite second moments suffice for this formula.⁸ The full-law result in Theorem 1 uses independence and transform admissibility as well.

It is worth keeping the two notions of order separate. Interaction order $|A|$ counts the policy switches in a component; cumulant order r counts the arguments of a probabilistic summary. A component involving many switches can still enter a second-order covariance.

6.2. Second-order marginal compatibility

For a set function $\nu : \mathcal{W} \rightarrow \mathbb{R}$, define the meet-kernel matrix $K_\nu \in \mathbb{R}^{n \times n}$, $n = 2^p$, by

$$(24) \quad (K_\nu)_{S,T} = \nu(S \cap T).$$

Let $\lambda = M\nu$ denote its Möbius transform, so

$$(25) \quad \lambda_A = \sum_{B \subseteq A} (-1)^{|A|-|B|} \nu(B).$$

PROPOSITION 3 (Meet-kernel factorization). *For every set function ν ,*

$$(26) \quad K_\nu = Z \text{diag}(\lambda_A) Z^\top.$$

Hence K_ν is positive semidefinite if and only if $\lambda_A \geq 0$ for every A . When this holds,

$$(27) \quad \text{rank}(K_\nu) = \#\{A : \lambda_A > 0\}.$$

PROOF. The (S, T) entry of the right side of (26) is

$$\sum_{A \subseteq S \cap T} \lambda_A = \nu(S \cap T)$$

⁸For a jointly Gaussian component vector, pairwise orthogonality implies mutual independence. Outside the Gaussian class, the variance identity characterizes second-order behavior; recovery of the full joint law uses additional restrictions.

by Möbius inversion. Since Z is invertible, congruence by Z preserves inertia. The positive-semidefinite and rank statements follow from the diagonal middle matrix. $\square$

Intersection covariance kernels have a long history in set-indexed Gaussian process theory; for example, Ossiander and Pyke (1985) study additive Gaussian processes with covariance $\mu(A \cap B)$. Nonnegative Möbius transforms also appear in the theory of completely monotone capacities (Chateauneuf and Jaffray 1989). For a surface compatible with the orthogonal benchmark, $\lambda_A \geq 0$ and these coefficients are component variances. A general variance surface obeys the union identity in Proposition 6.

PROPOSITION 4 (Variance-surface characterization). *Let $v(S) = \text{Var}(Y_S)$ for a square-integrable scalar system. Cofactual orthogonality implies*

$$(28) \quad \sigma_A^2 = (Mv)_A = \text{Var}(C_A) \geq 0$$

for every A . Conversely, if a proposed variance surface v satisfies $Mv \geq 0$, then there exists a centered cofactually factorized Gaussian system with $\text{Var}(Y_S) = v(S)$ for every S .

PROOF. Under cofactual orthogonality, variance additivity gives

$$v(S) = \sum_{A \in S} \text{Var}(C_A),$$

so Möbius inversion gives (28). For the converse, choose independent $C_A \sim N(0, \sigma_A^2)$ and set $Y_S = \sum_{A \in S} C_A$. $\square$

The signs of the coefficients have a population interpretation. A negative coefficient rules out every orthogonal coupling with those variances; nonnegative coefficients admit a Gaussian construction with the proposed variances. For specified non-Gaussian marginal laws, compatibility depends on the full CGF condition in Theorem 1.

In separate samples, $M\hat{v}$ estimates these population contrasts under the sampling conditions for the regime variances. Near a zero coefficient, statistical evidence depends on calibration at the boundary of the inequality cone (Andrews and Soares 2010). The paired equality tests use a different sampling experiment and calibration.

6.3. Common pre-intervention states

Shared observed heterogeneity can account for some cross-world dependence. Let X be an observed pre-intervention state in a standard Borel space, with the same interpretation in every regime. The incidence matrix remains Z . The conditional statements below hold for P_X -almost every x .

Write

$$\mu_A(x) = \mathbb{E}[C_A \mid X = x], \quad R_A = C_A - \mu_A(X), \quad \Omega_C(x) = \text{Cov}(C \mid X = x).$$

Conditional factorization means mutual independence of the C_A given X . Conditional orthogonality means that $\Omega_C(x)$ is diagonal. Applying Theorem 2 to each conditional law gives the corresponding finite-order cumulant hierarchy.

PROPOSITION 5 (Conditional calculus). *Under conditional orthogonality and finite second moments,*

$$(29) \quad \text{Cov}(Y_S, Y_T \mid X) = \text{Var}(Y_{S \cap T} \mid X).$$

Consequently,

$$(30) \quad \begin{aligned} \text{Cov}(Y_S, Y_T) &= \mathbb{E}[\text{Var}(Y_{S \cap T} \mid X)] \\ &+ \text{Cov}(\mathbb{E}[Y_S \mid X], \mathbb{E}[Y_T \mid X]). \end{aligned}$$

If the components are conditionally independent and the conditional joint MGF is finite on a neighborhood of zero, then

$$(31) \quad H_A(t \mid x) = \sum_{B \subseteq A} (-1)^{|A|-|B|} K_B(t \mid x),$$

and the conditional version of (10) determines the cross-world law given x . Knowledge of these conditional marginal laws and P_X therefore determines the unconditional joint law within this class.

Proof. See Appendix C.5.

Residualization removes the covariance due to variation in conditional means. The average residual product therefore measures average within-state covariance. That average can be zero when positive and negative conditional covariances offset each other. The selected state weights determine which of these patterns a test can detect.

6.4. What marginal cumulants aggregate

There is a useful identity that holds for every cross-world coupling. Let $k_r(S \mid x) = \kappa_r(Y_S \mid X = x)$ and assume conditional moments through order r exist.

PROPOSITION 6 (Union aggregation). *For every $B \subseteq [p]$,*

$$(32) \quad (Mk_r)_B(x) = \sum_{A_1 \cup \dots \cup A_r = B} \text{Cum}(C_{A_1}, \dots, C_{A_r} \mid X = x),$$

where the sum is over ordered tuples. In particular, if $v_S(x) = \text{Var}(Y_S \mid X = x)$, then

$$(33) \quad (Mv)_B(x) = \text{Var}(C_B \mid X = x) + \sum_{\substack{A \cup D = B \\ A \neq D}} \text{Cov}(C_A, C_D \mid X = x).$$

The same identities hold unconditionally when the corresponding moments exist.

Proof. See Appendix C.6.

The off-diagonal sum explains how a marginal variance contrast differs from the corresponding component variance. Orthogonality sets this correction to zero. Under a general coupling it can have either sign. For one intervention, it is twice the covariance between the baseline and the increment, as in the reward-credit calculation.

6.5. Variance error under component dependence

A policy-contrast variance can be compared with the diagonal-component benchmark. Let $\Omega_C(x) = \text{Cov}(C \mid X = x)$, and define

$$D(x) = \text{diag}\{\Omega_C(x)\}, \quad E(x) = \Omega_C(x) - D(x), \quad \eta_2 = \{\mathbb{E}\|E(X)\|_F^2\}^{1/2}.$$

Here $D(x)$ retains the actual conditional component variances. For a fixed world contrast b , set $w = Z^\top b$, so $b^\top Y = w^\top C$.

PROPOSITION 7 (Variance error). *Suppose $\mathbb{E}\|C\|^2 < \infty$ and $\eta_2 < \infty$. For every fixed $b \in \mathbb{R}^{\mathcal{W}}$,*

$$(34) \quad |\mathbb{E}[\text{Var}(b^\top Y \mid X)] - \mathbb{E}[w^\top D(X)w]| \leq \|w\|_2^2 \eta_2, \quad w = Z^\top b.$$

Proof. See Appendix C.7.

For a gain $Y_S - Y_T$, b places +1 and −1 at the two regimes. The bound holds the actual conditional component variances fixed and controls the contribution of their off-diagonal covariance. A marginal variance contrast has the additional interpretation given by the union correction in (33). Applying the bound therefore requires paired component information or a stated sensitivity restriction on η_2 .

For bounded functionals, a relative-entropy comparison can use the original law and a conditional product reference. The reference preserves the conditional component marginals. Its implied world marginals can change, so the comparison specifies both the reference law and the information used to bound the departure.

7. Observation and assessment

The coupling restriction specifies a class of joint laws. The observation scheme determines what can be learned about a law in that class. Table 2 distinguishes marginal samples, paired rows, and paired rows with a common state. The sampling results evaluate selected equalities for paired data; the acquisition problem adds a cost to those observations.

TABLE 2. Observation schemes and the quantities they support

Observation scheme	Information supplied	Analysis in this paper
Separate regime samples	Marginal laws or selected marginal moments	Population compatibility and coupling bounds; distributional sampling theory is a separate task.
Complete paired rows	A joint world vector for each common unit or draw	Covariance restrictions for the recorded joint law.
Common-state paired rows	Joint outcomes and a common pre-intervention state	Selected residual covariance moments with stated learner conditions.

Separate samples require their own assignment and support assumptions to identify regime marginals. Paired simulation provides the joint law specified by the simulator. Common-state adjustment uses the same pre-intervention interpretation of the state in every regime.

7.1. Observation fibers

Fix a class of candidate joint laws for the stacked world vector. All fibers in this section are taken within that class. Let Q denote one candidate law, and let

$$(35) \quad \mathcal{O} : Q \mapsto d$$

map latent laws to the observed object.

DEFINITION 6 (Observation fiber). *The fiber at d is*

$$(36) \quad \mathcal{F}(d) = \{Q : \mathcal{O}(Q) = d\}.$$

A functional is point identified at a feasible d when every law in $\mathcal{F}(d)$ gives it the same value. If this holds on each nonempty fiber, Proposition A7 writes the functional as a map of the observed object. This is a set-theoretic statement; regularity of the induced map is a further question. Changing the maintained class can change the fiber even with the observation map held fixed.

7.2. World marginals: unrestricted benchmark

Suppose the data identify every world marginal law

$$P = (P_S)_{S \in \mathcal{W}}, \quad P_S = \mathcal{L}(Y_S).$$

Under the unrestricted cross-world class, the compatible joint laws form the coupling set

$$\Pi(P).$$

This is the standard missing-cross-world problem in potential-outcome analysis. Distributional treatment-effect functionals can remain partially identified even when the marginals are well learned (Fan and Park 2009; Firpo and Ridder 2019; Frandsen and Lefgren 2021). Gao et al. (2025) formulate the general causal partial-identification problem directly as multi-marginal optimal transport and obtain the exact identified set together with statistical guarantees.

For a bounded continuous cofactual functional $h(MY)$ on a finite product of Euclidean outcome spaces, the same benchmark takes the form

$$(37) \quad \left[\inf_{\pi \in \Pi(P)} \int h(My) d\pi(y), \quad \sup_{\pi \in \Pi(P)} \int h(My) d\pi(y) \right].$$

The coupling set is weakly compact and convex, and the objective is continuous and affine, so the interval is attained and sharp. The Möbius transform changes the cost; the unresolved object is the cross-world coupling.

7.3. Separate-regime compatibility and its sampling scope

With compatible marginal laws, Corollary 1 leaves one joint law in the factorized fiber. The unrestricted fiber contains the coupling set above. Restricting only finitely many cumulant orders identifies the corresponding joint cumulants, with higher-order features still open. For selected moment observations, Proposition A8 in Appendix D gives the criterion in terms of feasible differences between observationally equivalent moment vectors.

At second order, the population object is $\lambda = Mv$, with compatibility requiring $\lambda_A \geq 0$ for every A . Separate-sample variance estimates give $\widehat{\lambda} = M\widehat{v}$. A sample coefficient close to zero needs to be read together with its sampling uncertainty. Joint inference must account for the dependence among the estimated contrasts and the boundary of the inequality cone (Andrews and Soares 2010). The formal tests below concern paired covariance equalities. Their guarantees apply to the recorded joint law under the stated sampling conditions.

From the policy target to a diagnostic. In (1), the pairing model supplies the cross-world covariance used to calculate gain dispersion. Paired observations allow assessment of restrictions on that covariance. With a common state, residual products isolate its within-state part. This fixes the diagnostic before choosing the sampling procedure.

7.4. Testing second-order restrictions

Each paired row records outcomes for one common unit or structural draw. It is the independent sampling unit in the results below. In a simulator, the shared draw implements the specified economic coupling. A repeated-regime application requires a stable unit-level outcome map and control of carryover effects. The sampling theory takes the pairing and structural parameters as fixed. With independent simulator draws, its standard errors describe Monte Carlo variation around moments of that fixed simulator law. In an empirical paired sample, they describe sampling variation under the stated unit-level observation scheme. Population gain dispersion in equation (1) is the economic target in both cases.⁹

Transforming a row by M gives the components. Their off-diagonal covariances are the moments under assessment. The null sets these moments to zero, and the covariance of their influence vectors supplies the Wald weighting matrix. This keeps the restriction on the outcome law separate from the variance used for inference.

Let $n = 2^p$ and $q = n(n-1)/2$. For a symmetric $n \times n$ matrix H , let $\text{vecoff}(H) \in \mathbb{R}^q$ stack the strict upper-triangular entries once. Given a paired draw Y , define

$$C = MY, \quad \mu_C = \mathbb{E}[C], \quad \Omega_C = \text{Cov}(C),$$

and write

$$(38) \quad g_2 = \text{vecoff}(\Omega_C).$$

Second-order cofactual orthogonality is the restriction $g_2 = 0$. With N i.i.d. paired draws, let $\widehat{\Omega}_C = N^{-1} \sum_j (C^{(j)} - \bar{C})(C^{(j)} - \bar{C})^\top$ be the sample covariance of the transformed draws $C^{(j)} = MY^{(j)}$, and set

$$\widehat{g}_2 = \text{vecoff}(\widehat{\Omega}_C).$$

For $z = C - \mu_C$, define the influence vector

$$(39) \quad \psi_2(C) = \text{vecoff}(zz^\top - \Omega_C), \quad V_2 = \mathbb{E}[\psi_2(C)\psi_2(C)^\top].$$

⁹Estimated structural primitives add a first-stage term to the influence expansion.

A sample analogue replaces μ_C and Ω_C by their empirical counterparts. All empirical covariance matrices below use denominator N .

THEOREM 3 (Cofactual Wald test). *Fix p as $N \rightarrow \infty$. Suppose $Y^{(1)}, \dots, Y^{(N)}$ are i.i.d. paired world draws, $\mathbb{E}\|Y\|^4 < \infty$, and V_2 is positive definite. Let $\widehat{V}_2$ be the empirical covariance of*

$$\widehat{\Psi}_2^{(j)} = \text{vecoff} \left[(C^{(j)} - \bar{C})(C^{(j)} - \bar{C})^\top - \widehat{\Omega}_C \right].$$

Under second-order cofactual orthogonality,

$$(40) \quad T_{2,N} = N \widehat{g}_2^\top \widehat{V}_2^{-1} \widehat{g}_2 \implies \chi_q^2.$$

Under a fixed alternative with $g_2 \neq 0$ and V_2 positive definite,

$$(41) \quad \frac{T_{2,N}}{N} \xrightarrow{P} g_2^\top V_2^{-1} g_2 > 0.$$

Hence the test that rejects for large values of $T_{2,N}$ is consistent.

Proof. See Appendix J.3.

This is a moment-restriction calculation (Hansen 1982; Newey and McFadden 1994) for the covariances implied by the fixed loading matrix. A rejection provides evidence against second-order orthogonality of the sampled coupling. The conclusion concerns those covariance moments. Assessment of higher-order dependence uses additional restrictions from the hierarchy.

Under independent components, the weighting matrix is a function of the component variances. A squared-correlation statistic is then available under stronger calibration conditions.

The local-power calculation and descriptive covariance projections use the same zero set as the covariance transform, with different weights on departures. Projection onto the linear meet space also differs from projection onto its positive-semidefinite covariance cone.

Scope of the calibration. The stated limit is pointwise. The law and intervention dimension are held fixed as the sample grows. The positive-definiteness condition on V_2 matters: deterministic components or redundant restrictions can give a singular influence covariance. Uniform calibration along sequences approaching singularity requires additional control. A prespecified subvector with positive-definite covariance has a chi-squared limit with degrees of freedom equal to its length; selecting the rank from the data requires its own calibration. The results condition on fixed structural parameters and a fixed latent pairing. An application with estimated primitives, serial dependence, or additional simulation error needs an influence expansion that includes those sources of uncertainty.

Orthogonality and the product shortcut. Let $U \sim N(0, 1)$, $C_\emptyset = U$, and $C_1 = U^2 - 1$. Then

$$\text{Cov}(C_\emptyset, C_1) = 0, \quad \text{Cum}(C_\emptyset, C_\emptyset, C_1) = 2.$$

The centered cross-product has variance $15 - 2 \cdot 3 + 1 = 10$. Multiplying the component variances gives $1 \cdot 2 = 2$. The robust statistic estimates the cross-product variance, which is the quantity needed for its standard error. The shortcut in Corollary A1 uses independence to replace it by the product. Simulations compare the two calibrations under this law and under independent and correlated laws.

7.5. State-adjusted moments and efficient inference

Fix a finite collection $\mathcal{J} = \{(A_\ell, B_\ell, h_\ell) : 1 \leq \ell \leq L\}$, where $A_\ell \neq B_\ell$ and each h_ℓ is a known bounded function of X . Define

$$(42) \quad H_\ell(X, C) = h_\ell(X)R_{A_\ell}R_{B_\ell}, \quad \theta_\ell = \mathbb{E}[H_\ell(X, C)].$$

Each instrument h_ℓ chooses a weighted average of a conditional component covariance. Conditional orthogonality sets every such moment to zero (Shah and Peters 2020; Scheidegger et al. 2022).

Use a fixed number of folds, with $\hat{\mu}_A^{(-j)}$ trained outside observation j 's fold. Set

$$(43) \quad \hat{\theta}_\ell = \frac{1}{N} \sum_{j=1}^N h_\ell(X_j) \{C_{A_\ell, j} - \hat{\mu}_{A_\ell}^{(-j)}(X_j)\} \{C_{B_\ell, j} - \hat{\mu}_{B_\ell}^{(-j)}(X_j)\}.$$

The implementation follows the observation design. Form the selected components from each paired row, train their conditional means on the other folds, and average the held-out residual products with the declared instruments. Each instrument identifies the state-weighted covariance question being assessed. Let $\hat{V}_X$ be the empirical covariance of the summand vectors $\hat{H}_j$.

Conditional centering makes the target locally insensitive to errors in either conditional mean. The L^2 product rate controls bias, and L^4 consistency controls variation in the estimated signal. Under the full conditions of Theorem A1, the influence function is $H - \theta$ and the estimator has expansion (A22). The empirical signal covariance estimates $V_X = \text{Var}(H)$, with a chi-squared calibration when it is positive definite. The efficiency bound is for these selected moments in the unrestricted paired-data model.

Scope of the null. The instrument collection determines which projections are tested. For example, let X be equally likely to be -1 or 1 , with $\text{Cov}(C_A, C_B \mid X) = \alpha X$. The instrument $h = 1$ gives $\theta = 0$ because the two states cancel. Using $h = X$ gives $\theta = \alpha$. All bounded measurable

instruments together characterize conditional covariance zero; a finite collection has the power scope of its selected projections. Conditional independence adds higher-order restrictions. Simulated paired worlds follow this example.

7.6. Finite-sample evidence

Table 3 brings together the two null designs used to assess the covariance calibration. Both satisfy second-order orthogonality; the dependent design has a different cross-product variance from the independence shortcut. The comparison therefore isolates the additional distributional condition needed by that shortcut. At $N = 250$, the robust rejection frequencies also show the approximation error of the chi-squared calibration.

TABLE 3. Calibration under two second-order nulls

N	Component law	Robust Wald (%)	Product shortcut (%)
250	Independent	6.95 (0.57)	5.00 (0.49)
250	Orthogonal, dependent	6.60 (0.56)	25.25 (0.97)
1,000	Independent	4.70 (0.47)	4.75 (0.48)
1,000	Orthogonal, dependent	4.90 (0.48)	24.25 (0.96)

Two interventions and 2,000 replications per row; nominal level 5%. Parentheses give Monte Carlo standard errors in percentage points. The independent design uses standard normal components. The dependent design sets $C_0 = U$ and $C_1 = (U^2 - 1)/\sqrt{2}$, with U standard normal and the other components independent standard normals. Both laws satisfy the covariance null. The product shortcut is calibrated under independence. These rows use the same rejection counts as Table A1, displayed here in percent.

The remaining simulations vary the intervention dimension and evaluate power against the two correlated alternatives. Their outputs have the scope of those moment tests.

The conditional experiments return to the sign-changing covariance example. Its unweighted projection is zero, whereas a state-weighted projection retains the dependence. The two tests therefore answer different moment questions. The reported rejection frequencies describe these particular implementations and sampling models.

8. Conclusion

A shared-component model links marginal regime laws to the distribution of policy gains. Under local exponential integrability, the Möbius-transformed marginal CGFs are admissible if and only if the independent-component representation exists. Compatible marginals determine its joint law. Gain comparisons cancel common components and combine the remaining laws.

The observations determine what can be assessed. Marginal variances give compatibility inequalities; paired rows permit tests of selected covariance equalities. With a

common pre-intervention state, residual moments assess within-state covariance. The formal sampling guarantees are pointwise for the selected paired moments under the stated fixed-dimensional conditions. The union identity and variance-error bound explain how component dependence affects the variance calculation.

The Bellman examples connect the restriction to economic primitives. A common evaluation operator preserves module independence; reoptimization can change it. The reward-credit example brings the maintained coupling, state adjustment, and the variance diagnostic into the same calculation. The resulting policy-gain law concerns outcomes joined for the same unit, type, or structural draw.

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